

Supporting information for: New Soft-Core Potential Function for Molecular Dynamics Based Alchemical Free Energy Calculations

Vytautas Gapsys,[†] Daniel Seeliger,^{†,‡} and Bert L. de Groot^{*,†}

*Computational biomolecular dynamics group, Max Planck Institute for biophysical Chemistry,
Göttingen, Germany*

E-mail: bgroot@gwdg.de

*To whom correspondence should be addressed

[†]Max Planck Institute for biophysical Chemistry

[‡]Department of Lead Identification and Optimization Support Boehringer Ingelheim Pharma GmbH Birkendorfer
Strasse 65 88397 Biberach, Germany

1 S1

Explicit expressions for the forces, potential energies and $\partial H/\partial \lambda$ curves. In all the following cases pairwise interactions between particles i and j are considered. For the sake of simplicity indices i and j were omitted where possible.

Lennard-Jones force:

$$\mathbf{F}_{LJ}(\mathbf{r}) = \begin{cases} \left(\frac{12C^{(12)}}{r^{13}} - \frac{6C^{(6)}}{r^7} \right) \frac{\mathbf{r}}{r}, & \text{if } r \geq r_{LJ} \\ \left(-\frac{156C^{(12)}}{r_{LJ}^{14}} + \frac{42C^{(6)}}{r_{LJ}^8} \right) \mathbf{r} + \frac{168C^{(12)}}{r_{LJ}^{13}} - \frac{48C^{(6)}}{r_{LJ}^7}, & \text{if } r < r_{LJ} \end{cases} \quad (1.1)$$

Coulomb force:

$$\mathbf{F}_Q(\mathbf{r}) = \begin{cases} \frac{q_i q_j}{4\pi\epsilon_0\epsilon_r r^2} \frac{\mathbf{r}}{r}, & \text{if } r \geq r_Q \\ -\frac{2q_i q_j}{r_Q^3} \mathbf{r} + \frac{3q_i q_j}{r_Q^2}, & \text{if } r < r_Q \end{cases} \quad (1.2)$$

Lennard-Jones potential energy:

$$V_{LJ}(r) = \begin{cases} \frac{C^{(12)}}{r^{12}} - \frac{C^{(6)}}{r^6}, & \text{if } r \geq r_{LJ} \\ \left(\frac{78C^{(12)}}{r_{LJ}^{14}} - \frac{21C^{(6)}}{r_{LJ}^8} \right) r^2 - \left(\frac{168C^{(12)}}{r_{LJ}^{13}} - \frac{48C^{(12)}}{r_{LJ}^7} \right) r + \frac{91C^{(12)}}{r_{LJ}^{12}} - \frac{28C^{(6)}}{r_{LJ}^6}, & \text{if } r < r_{LJ} \end{cases} \quad (1.3)$$

Coulomb potential energy:

$$V_Q(r) = \begin{cases} \frac{q_i q_j}{4\pi\epsilon_0\epsilon_r r}, & \text{if } r \geq r_Q \\ \frac{q_i q_j}{r_Q^3} r^2 - \frac{3q_i q_j}{r_Q^2} r + \frac{3q_i q_j}{r_Q}, & \text{if } r < r_Q \end{cases} \quad (1.4)$$

$\partial H/\partial \lambda$ for Lennard-Jones potential, when $r < r_{LJ}$:

$$\begin{aligned}
\frac{\partial H}{\partial \lambda} &= V_{LJ}^B(r) - V_{LJ}^A(r) + (1 - \lambda) \frac{\partial V_{LJ}^A(r)}{\partial \lambda} + \lambda \frac{\partial V_{LJ}^B(r)}{\partial \lambda} \\
&= \left(\frac{78C_B^{(12)}}{r_{LJB}^{14}} - \frac{21C_B^{(6)}}{r_{LJB}^8} \right) r^2 - \left(\frac{168C_B^{(12)}}{r_{LJB}^{13}} - \frac{48C_B^{(12)}}{r_{LJB}^7} \right) r + \frac{91C_B^{(12)}}{r_{LJB}^{12}} - \frac{28C_B^{(6)}}{r_{LJB}^6} \\
&\quad - \left(\frac{78C_A^{(12)}}{r_{LJA}^{14}} - \frac{21C_A^{(6)}}{r_{LJA}^8} \right) r^2 - \left(\frac{168C_A^{(12)}}{r_{LJA}^{13}} - \frac{48C_A^{(12)}}{r_{LJA}^7} \right) r + \frac{91C_A^{(12)}}{r_{LJA}^{12}} - \frac{28C_A^{(6)}}{r_{LJA}^6} \\
&\quad + \frac{14(\lambda - 1)}{\lambda} \left[\left(\frac{13C_A^{(12)}}{r_{LJA}^{14}} - \frac{2C_A^{(6)}}{r_{LJA}^8} \right) r^2 - \left(\frac{26C_A^{(12)}}{r_{LJA}^{13}} - \frac{4C_A^{(6)}}{r_{LJA}^7} \right) r + \frac{13C_A^{(12)}}{r_{LJA}^{12}} - \frac{2C_A^{(6)}}{r_{LJA}^6} \right] \\
&\quad + \frac{14\lambda}{1 - \lambda} \left[\left(\frac{13C_B^{(12)}}{r_{LJB}^{14}} - \frac{2C_B^{(6)}}{r_{LJB}^8} \right) r^2 - \left(\frac{26C_B^{(12)}}{r_{LJB}^{13}} - \frac{4C_B^{(6)}}{r_{LJB}^7} \right) r + \frac{13C_B^{(12)}}{r_{LJB}^{12}} - \frac{2C_B^{(6)}}{r_{LJB}^6} \right]
\end{aligned} \tag{1.5}$$

$\partial H/\partial \lambda$ for Coulomb potential, when $r < r_Q$:

$$\begin{aligned}
\frac{\partial H}{\partial \lambda} &= V_Q^B(r) - V_Q^A(r) + (1 - \lambda) \frac{\partial V_Q^A(r)}{\partial \lambda} + \lambda \frac{\partial V_Q^B(r)}{\partial \lambda} \\
&= \frac{q_i^B q_j^B}{r_{QB}^3} r^2 - \frac{3q_i^B q_j^B}{r_{QB}^2} r + \frac{3q_i^B q_j^B}{r_{QB}} \\
&\quad - \left(\frac{q_i^A q_j^A}{r_{QA}^3} r^2 - \frac{3q_i^A q_j^A}{r_{QA}^2} r + \frac{3q_i^A q_j^A}{r_{QA}} \right) \\
&\quad + \frac{\lambda - 1}{2\lambda} \left(\frac{q_i^A q_j^A}{r_{QA}^3} r^2 - \frac{2q_i^A q_j^A}{r_{QA}^2} r + \frac{q_i^A q_j^A}{r_{QA}} \right) \\
&\quad + \frac{\lambda}{2(1 - \lambda)} \left(\frac{q_i^B q_j^B}{r_{QB}^3} r^2 - \frac{2q_i^B q_j^B}{r_{QB}^2} r + \frac{q_i^B q_j^B}{r_{QB}} \right)
\end{aligned} \tag{1.6}$$

2 S2

Here we show that both the force and potential energy functions at the switching point for the new soft-core function converge to the original, unperturbed (“hard-core”) formulation of the Lennard-Jones and Coulomb interactions, respectively.

Continuity of the Lennard-Jones force:

$$\lim_{r \rightarrow r_{LJ}} \left(-\frac{156C^{(12)}}{r_{LJ}^{14}} + \frac{42C^{(6)}}{r_{LJ}^8} \right) \mathbf{r} + \frac{168C^{(12)}}{r_{LJ}^{13}} - \frac{48C^{(6)}}{r_{LJ}^7} = \left(\frac{12C^{(12)}}{r_{LJ}^{13}} - \frac{6C^{(6)}}{r_{LJ}^7} \right) \frac{\mathbf{r}_{LJ}}{r_{LJ}} \quad (2.1)$$

Continuity of the Coulomb force:

$$\lim_{r \rightarrow r_Q} \left(\frac{2q_i q_j}{r_Q^3} \mathbf{r} + \frac{3q_i q_j}{r_Q^2} \right) = \frac{q_i q_j}{4\pi\epsilon_0\epsilon_r r_Q^2} \frac{\mathbf{r}_Q}{r_Q} \quad (2.2)$$

Continuity of the Lennard-Jones energy:

$$\lim_{r \rightarrow r_Q} \left[\left(\frac{78C^{(12)}}{r_{LJ}^{14}} - \frac{21C^{(6)}}{r_{LJ}^8} \right) r^2 - \left(\frac{168C^{(12)}}{r_{LJ}^{13}} - \frac{48C^{(6)}}{r_{LJ}^7} \right) r + \frac{91C^{(12)}}{r_{LJ}^{12}} - \frac{28C^{(6)}}{r_{LJ}^6} \right] = \frac{C^{(12)}}{r_{LJ}^{12}} - \frac{C^{(6)}}{r_{LJ}^6} \quad (2.3)$$

Continuity of the Coulomb energy:

$$\lim_{r \rightarrow r_Q} \left(\frac{q_i q_j}{r_Q^3} r^2 - \frac{3q_i q_j}{r_Q^2} r + \frac{3q_i q_j}{r_Q} \right) = \frac{q_i q_j}{4\pi\epsilon_0\epsilon_r r_Q} \quad (2.4)$$

3 S3

The derivation of $\partial H/\partial \lambda$.

The switch function can be written as a sum of two functions by employing the Heaviside step function. The force for the new soft-core function is given by:

$$\mathbf{F}(\lambda, \mathbf{r}) = H(r_0(\lambda) - r)\mathbf{f}_1(\lambda, \mathbf{r}) + H(r - r_0(\lambda))\mathbf{f}_2(\lambda, \mathbf{r}), \quad (3.1)$$

where H is a Heaviside function defined as $H(r) = \begin{cases} 0, & \text{if } x < 0 \\ \frac{1}{2}, & \text{if } x = 0. \\ 1, & \text{if } x > 0 \end{cases}$

$\mathbf{f}_1(\lambda, \mathbf{r})$ is the linearized force at close distances $r < r_0(\lambda)$, $\mathbf{f}_2(\lambda, \mathbf{r})$ is the usual expression of the Coulomb or Lennard-Jones force at distances $r \geq r_0(\lambda)$. At the point r_0 the classical (“hard-core”) description of the non-bonded interactions is used for the force, because $\mathbf{f}_1(\lambda, r_0) = \mathbf{f}_2(\lambda, r_0)$ (see S2 equations Eq. (2.1) and Eq. (2.2)). $\mathbf{f}(\lambda, \mathbf{r}) = (1 - \lambda)\mathbf{f}_A(\mathbf{r}) + \lambda\mathbf{f}_B(\mathbf{r})$.

The potential energy for the new soft-core function:

$$V(\lambda, r) = H(r_0(\lambda) - r)v_1(\lambda, r) + H(r - r_0(\lambda))v_2(\lambda, r), \quad (3.2)$$

where $v(\lambda, r) = -\int \mathbf{f}(\lambda, \mathbf{r})d\mathbf{r}$.

The derivative of the potential energy with respect to λ :

$$\begin{aligned} \frac{\partial V(\lambda, r)}{\partial \lambda} &= \frac{\partial H(r_0(\lambda) - r)}{\partial \lambda}v_1(\lambda, r) + H(r_0(\lambda) - r)\frac{\partial v_1(\lambda, r)}{\partial \lambda} \\ &\quad + \frac{\partial H(r - r_0(\lambda))}{\partial \lambda}v_2(\lambda, r) + H(r - r_0(\lambda))\frac{\partial v_2(\lambda, r)}{\partial \lambda} \\ &= -\delta(r - r_0(\lambda))\frac{dr_0(\lambda)}{d\lambda}v_1(\lambda, r) + H(r_0(\lambda) - r)\frac{\partial v_1(\lambda, r)}{\partial \lambda} \\ &\quad + \delta(r - r_0(\lambda))\frac{dr_0(\lambda)}{d\lambda}v_2(\lambda, r) + H(r - r_0(\lambda))\frac{\partial v_2(\lambda, r)}{\partial \lambda} \\ &= H(r_0(\lambda) - r)\frac{\partial v_1(\lambda, r)}{\partial \lambda} + H(r - r_0(\lambda))\frac{\partial v_2(\lambda, r)}{\partial \lambda} \end{aligned} \quad (3.3)$$

where $\delta(r)$ is Dirac's delta.

4 S4

Probability estimation of having zero distance between two particles.

The analytical solution for the nearest neighbor distance distribution in ideal gas can be found in.¹

The distribution is plotted in the Figure S1 (red curve) and its expression is the following

$$w(r) = e^{-4\pi r^3 n} 4\pi r^2 n, \quad (4.1)$$

where n is the number of particles per unit volume (N/V , N - total number of particles, V - volume).

Following from the expression, the probability to find two completely overlapping particles is zero.

The example of the ideal gas illustrates that even for particles without any interatomic potential, overlap (i.e. zero nearest neighbor distance) is entropically forbidden. In the simulations, however, complete overlap could occur due the finite machine precision. To analyse the nearest neighbor distance distribution, molecular dynamics simulations mimicking ideal gas were carried out: 212 particles were placed in a cubic box of volume 8 nm^3 periodically replicated in all directions. Lennard-Jones C_{12} and C_6 parameters and charges were set to 0 for all the particles. Stochastic dynamics simulation was performed for 250 ps. The minimal interatomic distances observed were compared with the theoretical distribution (Figure S1). The nearest neighbor distribution (black curve in Figure S1) at short distances converges to zero. The smallest NN distance observed in the simulation was $7.4 \times 10^{-4} \text{ nm}$. From the cumulative probabilities of finding two particles at short distances (Table S1) it can be deduced that reaching zero distance is highly unlikely due to the entropic barrier even at the single machine precision (~ 7 floating point digits), not to mention double precision (~ 16 floating point digits).

In simulations of biomolecules the distribution of the nearest neighbor distances between particles will be strongly influenced by the interaction potential between atoms. To have a hint on the smallest distances that could be observed in the simulation with interactions we performed two runs of MD using the setup described above, but assigning non-zero charges to the particles. In the first setup 106 molecules were assigned $+0.05$ and the remaining 106 particles got -0.05 partial charges,

whereas for the second setup we increased the charges to +0.1 and -0.1. The new soft-core potential was applied to avoid singularity points. In this case we observe a modified NN distance distribution (blue and green curves in Figure S1) with a shoulder towards the small distances. Nevertheless the distributions clearly go to zero when the inter-particle distances approach zero. The minimal observed values for the simulations with the charged particles were 4.9×10^{-4} nm and 7.0×10^{-4} for the 0.05/-0.05 and 0.1/-0.1 charges, respectively. Also the cumulative distributions of finding the particles at short distances rapidly converge to zero.

From this we conclude that even at the single machine precision entropic barrier is sufficiently large to never reach the zero distance between two particles. Hence, the discontinuity of the new soft-core function at zero inter-particle distance is not a problem for the practical application in the simulations.

It could be argued that atoms may be artificially set to a complete overlap in the beginning of a simulation. However, if the simulation starts at the $\lambda = 0$ or $\lambda = 1$ point the soft-core interaction plays no role. Another interesting observation concerning the discontinuity at zero inter-particle distance: the hard-core non-bonded interaction potentials also suffer from the same discontinuity at zero distance. In case two particles are artificially set to overlap completely (i.e. the distance between them is zero) and hard-core potential is used to describe their interaction, the spatial projections of the force will be zero, as it is for the new soft-core function.

Table S1: Probabilities of finding particles in a close distance region.

NN distance (nm)	Theoretical distribution	$q_1 = 0$ $q_2 = 0$	$q_1 = +0.05$ $q_2 = -0.05$	$q_1 = 0.1$ $q_2 = 0.1$
0.1	1.1×10^{-1}	1.0×10^{-1}	1.6×10^{-1}	2.1×10^{-1}
0.01	1.1×10^{-4}	1.2×10^{-4}	8.8×10^{-4}	6.1×10^{-4}
0.001	1.1×10^{-7}	1.6×10^{-7}	1.0×10^{-6}	3.2×10^{-7}
0.0001	1.1×10^{-10}	-	-	-
10^{-6}	1.1×10^{-16}	-	-	-
10^{-7}	1.1×10^{-19}	-	-	-
10^{-16}	1.1×10^{-46}	-	-	-

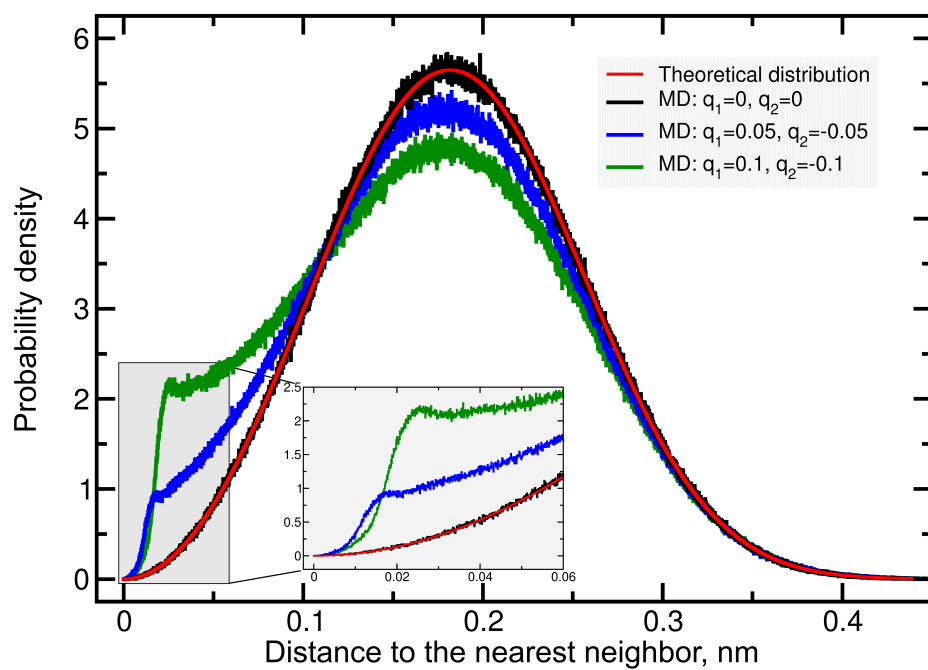


Figure S1: Nearest neighbor distance distributions: theoretical distribution (red curve), simulations with the charges set to zero (black), charges 0.05/-0.05 (blue), charges 0.1/-0.1 (green).

5 S5

Additional minima caused by the soft-core potential.

Occurrence of additional minima in the non-bonded pair-wise interactions can be observed by analysing the potential energy and force surfaces during an alchemical transition. Figure S2 shows the analytically calculated surfaces of non-bonded interaction (Coulomb and Lennard-Jones considered together) energy and force for two atoms. Parameters of the atoms and soft-core potential are identical to those given for Figure 2 in the main manuscript. States A and B for this alchemical transition are identical, thus the surfaces clearly show the effect of the soft-core potential. The distance between the atoms is depicted on the y axis, whereas x axis shows the λ value. In case of the hard-core potential the particles in close distance feel strong repulsive force for any state of λ . The conventional soft-core function has a broad region of zero or very small force for short interatomic distances throughout the transition. The force surface for the new soft-core function is more similar to the hard-core interaction surface. Figure 2 in the main manuscript shows the pair-wise interaction energy and force curves at $\lambda = 0.5$ for the same transition as represented by the surfaces in Figure S2. As discussed in the section 2.2 zero force at short distances may cause instabilities during the alchemical transitions.

To illustrate this we calculated the potential and force surfaces for 4 interacting atoms mimicking the situation depicted in Figure 1 in the manuscript. In the modeled situation two atoms (N and H_{N1}) are appearing in the system during the transition from state A to B. The other two atoms (H_{W1} and O_W) are present in the system and remain unchanged in both states. The geometry of the interaction is shown in the scheme in the Figure S3 A: a) N and H_{N1} are connected via a rigid bond; b) H_{W1} and O_W are connected via a rigid bond; c) the vectors N- H_{N1} and H_{W1} - O_W are parallel; d) atoms H_{N1} and H_{W1} are facing each other and the distance between them is plotted on the y axis of the Figure S3; e) a dummy site is bound to the O_W atom. The Lennard-Jones parameters, partial charges, bond lengths and angle between O_W and dummy were set to match the OPLS force field and TIP4P water parameters. In the Figure S3 B we observe constant repul-

sion at short distances when the hard-core potential is used. The conventional soft-core shows an attractive force at short distances. As λ value approaches 1, the minimum in the potential energy is shifting towards shorter distances (less than 0.1 nm). In the last steps of the transition, the additional minimum disappears and a strong repulsion between the particles is observed, as described in section 2.2 and Figure 1 of the main manuscript. The occurrence of this unwanted minimum is a result of an imbalance between a stronger electrostatic attraction and weaker Lennard-Jones repulsion arising between more than two atoms when their interactions are soft-cored following the Beutler et al.² formulation. The new soft-core potential avoids the occurrence of additional minima at short distances.

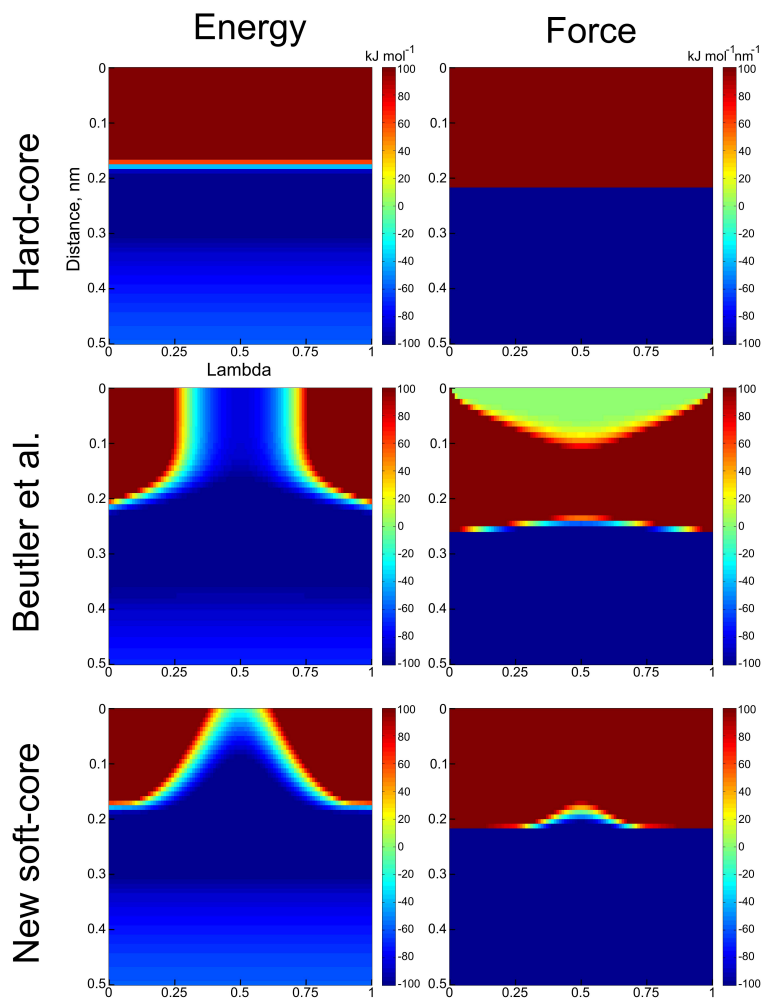


Figure S2: Interaction between 2 atoms. Potential energy (first column) and force (second column) surfaces for two interacting atoms. Both atoms under consideration were assigned identical Lennard-Jones parameters: $\sigma = 0.3$ nm and $\varepsilon = 0.5$ kJ/mol. Partial charges of the atoms were: $q_1 = 0.5$ and $q_2 = -0.5$. Gromacs 4.5 soft-core parameters: $\alpha = 0.3$ and $\sigma = 0.3$. Parameters for the new soft-core function: $\alpha_{LJ} = 0.85$, $\alpha_Q = 0.3$ and $\sigma_Q = 1$. The states A and B were identical.

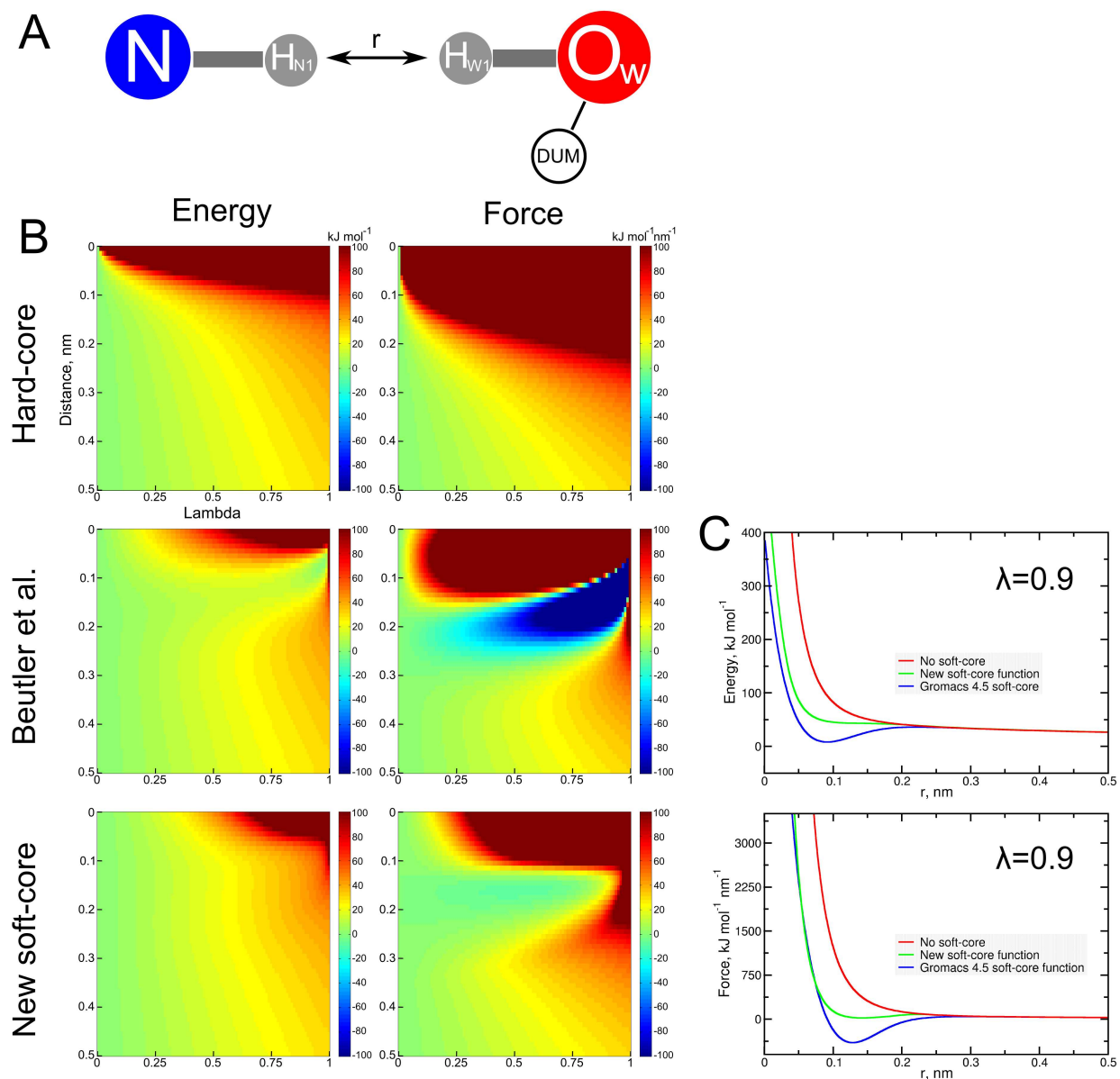


Figure S3: A: Schematic representation of the geometry of the 4 atom interaction used to calculate energy and force surfaces. B: Potential energy (first column) and force (second column) surfaces for 4 atom interaction. The parameters of the atoms were taken from the OPLS force field and TIP4P water model. Gromacs 4.5 soft-core parameters: $\alpha = 0.3$ and $\sigma = 0.3$. Parameters for the new soft-core function: $\alpha_{LJ} = 0.85$, $\alpha_Q = 0.3$ and $\sigma_Q = 1$. The states A and B were identical. C: A cut through the potential energy (top) and force (bottom) surfaces at $\lambda = 0.9$.

In Figure S4 we show that selecting different parameter sets for the original Beutler et al. soft-core function can significantly change the potential energy and force landscapes for the non-bonded interactions of two atoms. Our initial attempts aimed at selecting modified parameters to avoid local minima. However, introducing separate Lennard-Jones and Coulomb parameters does not alleviate the problem of additional minima.

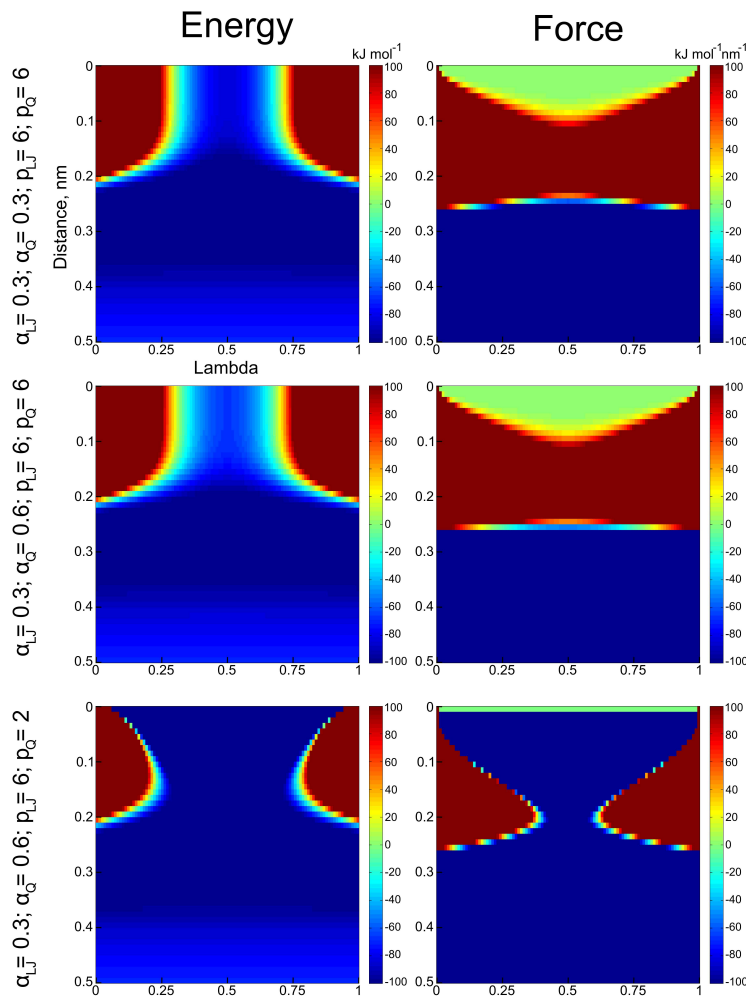


Figure S4: Influence of the parameters on the Lennard-Jones and Coulomb interactions for the Beutler et al. soft-core formulation as implemented in Gromacs 4.5. The interaction between 2 atoms was considered. Potential energy (first column) and force (second column) surfaces for two interacting atoms. Both atoms under consideration were assigned identical Lennard-Jones parameters: $\sigma = 0.3$ nm and $\varepsilon = 0.5$ kJ/mol. Partial charges of the atoms were: $q_1 = 0.5$ and $q_2 = -0.5$. The first row: $\alpha_{LJ} = 0.3$ and $\alpha_Q = 0.3$, $p_{LJ} = 6$ and $p_Q = 6$. The second row: $\alpha_{LJ} = 0.3$ and $\alpha_Q = 0.6$, $p_{LJ} = 6$ and $p_Q = 6$. The third row: $\alpha_{LJ} = 0.3$ and $\alpha_Q = 0.6$, $p_{LJ} = 6$ and $p_Q = 2$.

6 S6

Parameter selection.

The validity of the parameter set for the new soft-core function was assessed by analysing the potential energy and force surfaces for two interacting atoms. Here we chose two atoms with large partial charges of opposite sign, thus we could observe whether the soft-cored attractive electrostatic interaction would overcome the linearized Lennard-Jones repulsion. In our test we used Cl and Ca ions from the Amber99sb force field, carrying -1 and +2 charges, respectively. In Figure S5 we can see that with the current parameter set ($\alpha_{LJ} = 0.85$, $\alpha_Q = 0.3$, $\sigma_Q = 1$) no attractive force at short distances occurs. For a more complex analysis of more than 2 interacting atoms a simulation is needed to validate the parameter set. The simulations and their results are described in the main manuscript.

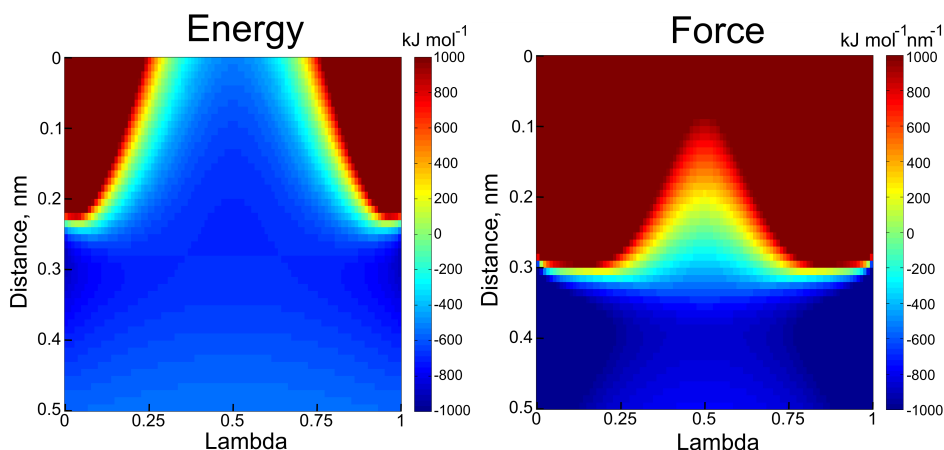


Figure S5: Force and potential curves for the non-bonded interactions between two atoms at $\lambda = 0.5$. Lennard-Jones parameters: $\sigma_{Cl} = 0.440104$ nm and $\epsilon_{Cl} = 0.4184$ kJ/mol, $\sigma_{Ca} = 0.30524$ nm and $\epsilon_{Ca} = 1.92376$ kJ/mol. Charges: $q_{Cl} = -1$ and $q_{Ca} = +2$. The states A and B were identical. Parameters for the new soft-core function: $\alpha_{LJ} = 0.85$, $\alpha_Q = 0.3$ and $\sigma_Q = 1$.

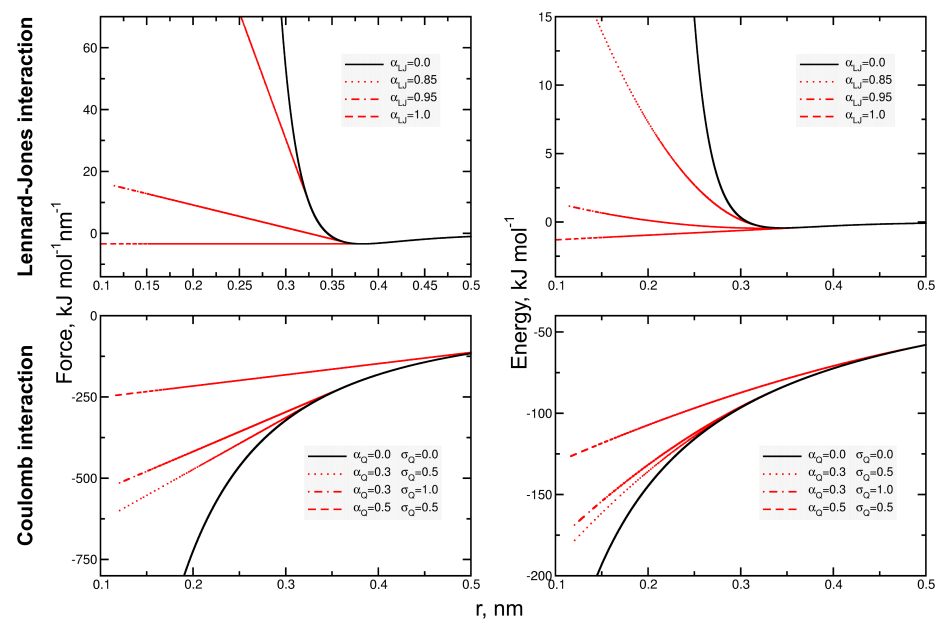


Figure S6: Dependence of the new soft-core force and energy curves on the different parameter sets used. Curves were generated by monitoring the interaction of a carbon atom (Lennard-Jones parameters $\sigma = 0.3$ nm, $\epsilon = 0.3$ kJ/mol, charge $q_C = 0.25$) with the TIP3P water oxygen (Lennard-Jones parameters $\sigma = 0.315061$ nm, $\epsilon = 0.636386$ kJ/mol, charge $q_O = -0.834$). The carbon atom was disappeared from the system and the forces and energies at the $\lambda = 0.5$ state were recorded.

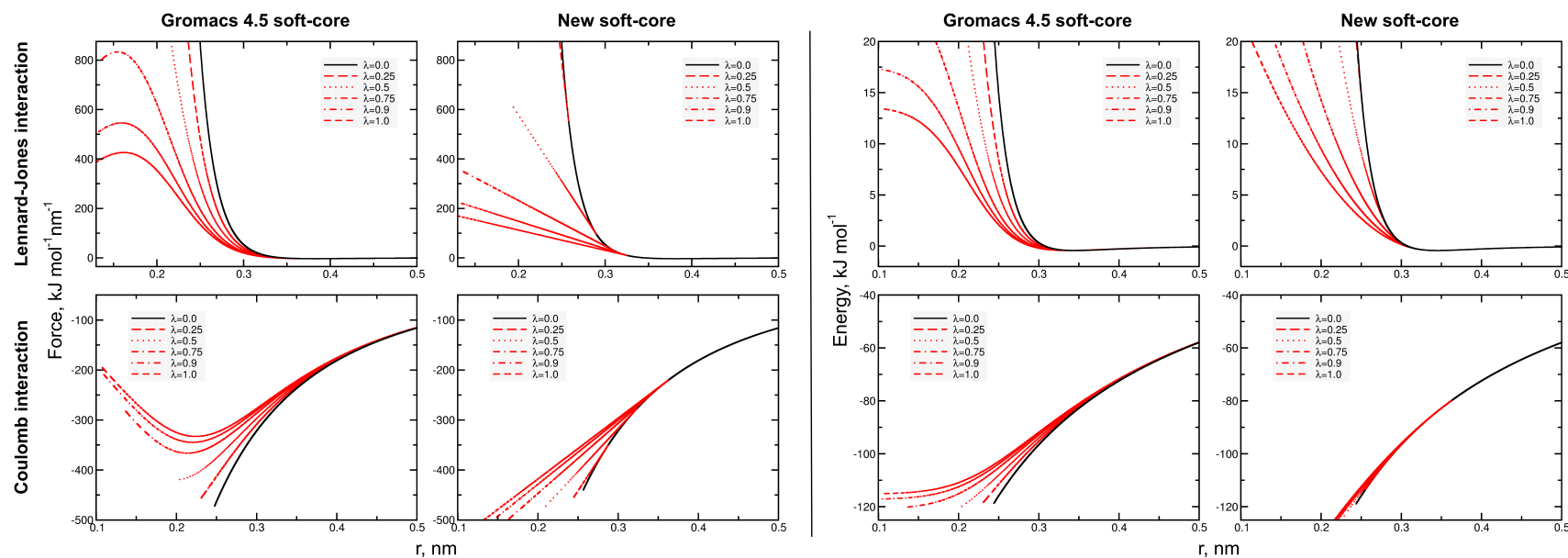


Figure S7: Comparison of the changes in the Gromacs 4.5 and new soft-core energy and force curves with the change in λ . Curves were generated by monitoring the interaction of a carbon atom (Lennard-Jones parameters $\sigma = 0.3$ nm, $\varepsilon = 0.3$ kJ/mol, charge $q_C = 0.25$) with the TIP3P water oxygen (Lennard-Jones parameters $\sigma = 0.315061$ nm, $\varepsilon = 0.636386$ kJ/mol, charge $q_O = -0.834$). The carbon atom was disappeared from the system during the transition. Parameters for the Gromacs 4.5 soft-core function: $\alpha = 0.3$, $\alpha = 0.25$ and $p = 1$. Parameters for the new soft-core function: $\alpha_{LJ} = 0.85$, $\alpha_Q = 0.3$ and $\sigma_Q = 1$.

9 S9

σ_Q parameter

Two main reasons guided introduction of the σ_Q parameter in the formulation of the new soft-core function. Firstly, σ_Q allows fine tuning of the linearization point of the electrostatic interaction. Figure S8 shows the effect of the changes in the σ_Q value. σ_Q parameter gives access to control the strength of the Coulomb component of the soft-core function based on the nature of interacting atoms, which was the second reason to introduce σ_Q .

In detail, the distance at which Lennard-Jones interaction is linearized ($r_{ij}^{LJ} = \alpha_{LJ}(\frac{26}{7} \frac{C_{ij}^{(12)}}{C_{ij}^{(6)}} \lambda)^{\frac{1}{6}}$) depends on the $C^{(12)}$, $C^{(6)}$ parameters, which define the position of the minimum of the Lennard-Jones potential and force. Since electrostatic interaction does not have a minimum, defining its soft-core linearization point is less strictly defined. To stay consistent with the soft-core for the Lennard-Jones interaction we construct the $r_{ij}^Q = (1 + \sigma_Q |q_i q_j|) \alpha_Q \lambda^{\frac{1}{6}}$ such that atoms with larger partial charges are "soft-cored" stronger. σ_Q allows us to switch on/off this charge dependence or modify its strength. This parameter is usually not of vital importance and can be kept at its default value for most practical purposes.

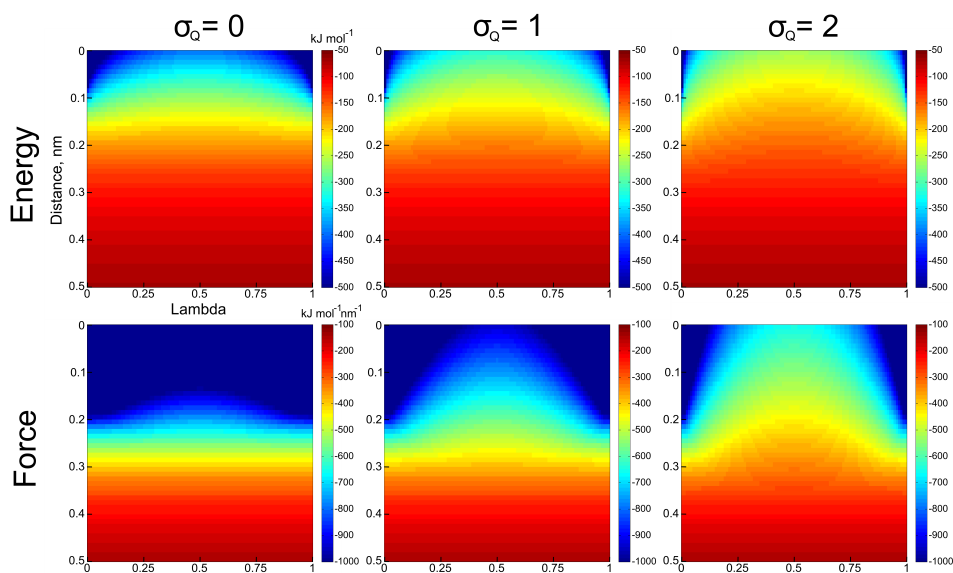


Figure S8: Dependence of the electrostatic potential with the new soft-core function on the value of σ_Q parameter. Potential energy (first row) and force (second row) surfaces for two atom interaction were calculated for three σ_Q values: 0 (first column), 1 (second column), 2 (third column). The partial charges of the atoms were set to $q_1 = 0.5$ and $q_2 = -0.5$. Soft-core parameter α_Q was set to 0.3.

Table S2: Barnase mutations. Free energy values with standard errors.

Mutation (kJ/mol)	Experiment ³ (kJ/mol)	New soft-core ffAmber99sb (kJ/mol)	Seeliger and de Groot ⁴ ffAmber99sb (kJ/mol)	New soft-core OPLS (kJ/mol)	Gromacs 4.5 soft-core OPLS (kJ/mol)
A30C	-4.18 ± 0.13	-2.18 ± 0.30	-1.57 ± 0.34	-3.38 ± 0.39	-4.71 ± 0.33
A30F	-2.93 ± 0.13	-4.29 ± 0.58	-2.70 ± 0.54	-8.42 ± 0.69	-7.94 ± 0.63
A30G	-3.76 ± 0.13	-3.10 ± 0.33	-3.00 ± 1.83	-11.91 ± 0.33	-12.13 ± 0.33
A30H	-3.34 ± 0.13	-0.93 ± 0.48	-2.25 ± 0.53	1.04 ± 0.72	1.81 ± 0.77
A30I	-3.34 ± 0.13	-3.43 ± 0.61	-3.90 ± 0.59	-1.27 ± 0.56	-0.98 ± 0.57
A30L	-1.25 ± 0.13	2.48 ± 0.48	3.51 ± 0.47	-2.44 ± 0.52	-2.21 ± 0.52
A30M	-1.25 ± 0.13	-0.78 ± 0.49	-1.21 ± 0.50	-1.00 ± 0.57	-1.12 ± 0.60
A30N	-2.93 ± 0.13	-1.01 ± 0.46	0.18 ± 0.74	-0.72 ± 0.50	0.53 ± 1.41
A30Q	-2.09 ± 0.13	-1.49 ± 0.53	-0.30 ± 0.65	-1.63 ± 0.71	-0.63 ± 0.86
A30S	-1.67 ± 0.13	-0.84 ± 0.37	0.14 ± 0.35	-1.79 ± 0.40	-2.04 ± 0.64
A30T	-3.34 ± 0.13	-0.96 ± 0.81	-0.78 ± 0.64	-8.85 ± 0.85	-10.39 ± 1.00
A30V	-3.76 ± 0.13	-4.44 ± 0.42	-3.95 ± 0.47	-4.77 ± 0.51	-4.30 ± 0.52
A30W	-4.18 ± 0.13	-2.87 ± 0.77	-2.32 ± 0.67	-4.38 ± 0.99	-4.72 ± 1.25
A30Y	-3.34 ± 0.13	-3.78 ± 0.60	-4.45 ± 0.57	-7.86 ± 0.74	-6.62 ± 0.94

Table S3: Solvation free energies. Free energy values with standard errors.

Ligand (kJ/mol)	Experiment (kJ/mol)	Nicholls et al. ⁵ (kJ/mol)	New soft-core (kJ/mol)	Gromacs 4.5 soft-core (kJ/mol)
triacetin	-46.90 ± 0.84^6	-50.88 ± 0.42	-45.26 ± 0.80	-46.29 ± 0.69
benzylbromide	$-5.90 \pm 0.84^{6,7}$	-4.64 ± 0.42	-5.20 ± 0.44	-5.62 ± 0.38
benzylchloride	$-6.23 \pm 0.84^{6,7}$	-6.28 ± 0.33	-6.40 ± 0.42	-5.26 ± 0.39
M-ditrifluoromethylbenzene	4.23 ± 0.84^6	-1.13 ± 0.17	2.44 ± 0.67	2.85 ± 0.49
N,N-Dimethyl-pmethoxybenzamide	-30.29 ± 0.84^8	-39.37 ± 0.25	-35.69 ± 0.66	-33.80 ± 0.52
N,N,4-trimethylbenzamide	-24.52 ± 0.84^8	-36.65 ± 0.21	-29.81 ± 0.60	-28.93 ± 0.52
chloroethyl ether	$-8.91 \pm 0.84^{6,7}$	-8.66 ± 0.04	-9.83 ± 0.51	-8.86 ± 0.43
1-acetyloxyethyl acetate	-29.96 ± 0.84^6	-33.64 ± 0.13	-28.42 ± 0.48	-28.36 ± 0.45
1,1-diethoxyethane	$-1.00 \pm 0.84^{6,9}$	-7.91 ± 0.33	-10.26 ± 0.54	-9.29 ± 0.46
dioxane	$-18.74 \pm 0.84^{6,9}$	-17.99 ± 0.13	-16.03 ± 0.32	-16.24 ± 0.31
diethyl propanedioate	-28.70 ± 0.84^6	-30.21 ± 0.25	-27.66 ± 0.61	-27.63 ± 0.53
dimethoxymethane	$-7.03 \pm 0.84^{6,9}$	-13.39 ± 0.42	-13.05 ± 0.37	-11.53 ± 0.36
2-acetyloxyethyl acetate	-32.22 ± 0.84^6	-35.19 ± 0.25	-32.65 ± 0.55	-32.10 ± 0.50
1,2-diethoxyethane	$-6.32 \pm 0.84^{6,9}$	-13.10 ± 0.08	-12.78 ± 0.53	-12.48 ± 0.47
ethylsulfanyethane	2.97 ± 0.84^{10}	1.34 ± 0.25	-1.29 ± 0.37	-1.29 ± 0.33
phenyl formate	-21.05 ± 0.84^5	-22.01 ± 0.13	-21.72 ± 0.43	-21.06 ± 0.36
imidazole	-34.64 ± 0.84^{11}	-32.84 ± 0.13	-34.03 ± 0.28	-33.59 ± 0.25

12 S12

Table S4: Free energies of the trypsin inhibitors. Free energy values with standard errors.

Ligand (kJ/mol)	Experiment ¹² (kJ/mol)	Talhout et al. ¹² (kJ/mol)	New soft-core (kJ/mol)	Gromacs 4.5 soft-core (kJ/mol)
Ben2Meth	-1.00 ± 0.20	-1.30 ± 0.80	1.73 ± 0.44	1.55 ± 1.16
Met2Eth	2.10 ± 0.10	0.90 ± 0.90	1.73 ± 0.40	-0.34 ± 0.95
Eth2Prp	-0.30 ± 0.30	-2.50 ± 0.90	0.93 ± 0.49	2.65 ± 1.05
nPrp2nBut	-0.50 ± 0.30	-1.60 ± 1.00	-1.89 ± 0.77	1.08 ± 1.18
nBut2nPent	-1.00 ± 0.10	-2.40 ± 0.70	-1.22 ± 1.17	-3.74 ± 1.42
nPent2nHex	-2.00 ± 0.10	-1.80 ± 0.90	-2.16 ± 1.39	-4.34 ± 1.64
Eth2iPrp	2.70 ± 0.30	1.10 ± 0.90	4.62 ± 0.47	4.24 ± 0.82

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